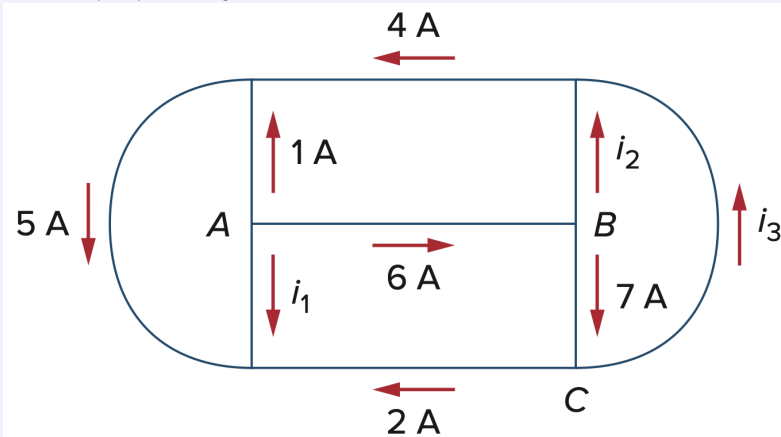


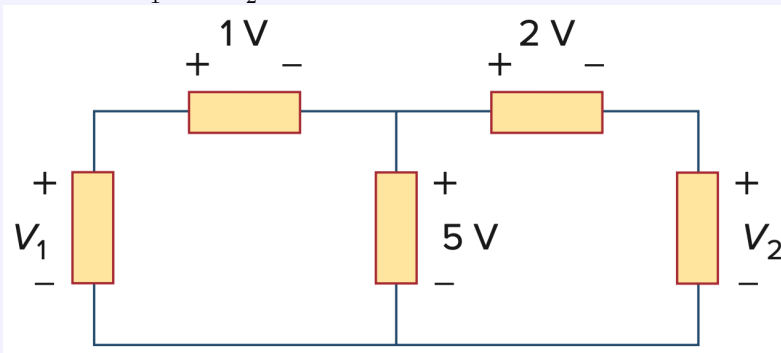
Exercise Sheet #2:

DC 1

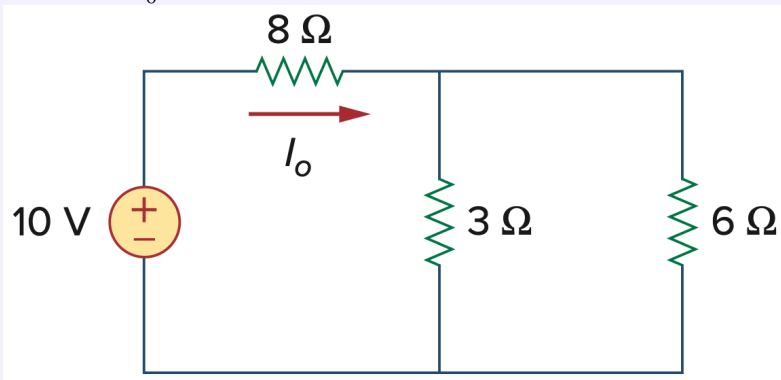
Exercise 1: KCL

Find i_1 , i_2 , and i_3 .

Exercise 2: KVL

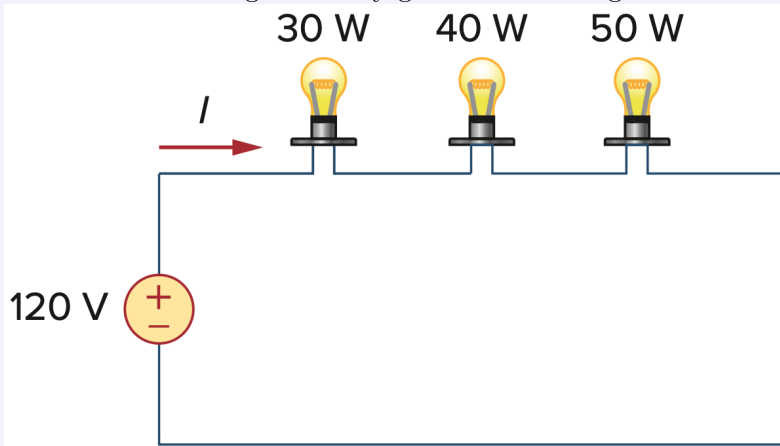
Calculate V_1 and V_2 .

Exercise 3: Series and Parallel Resistors

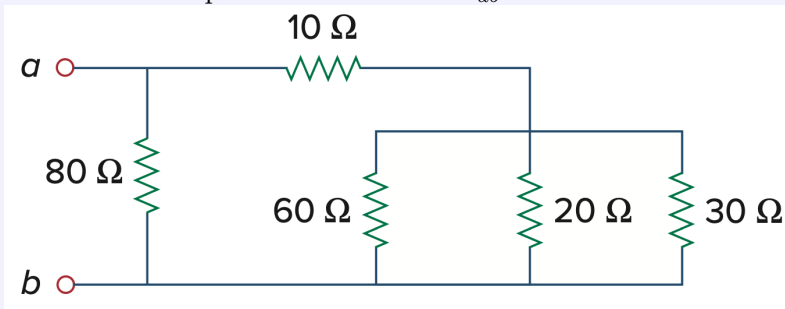
Calculate I_0 .

Exercise 4: Series vs. Parallel

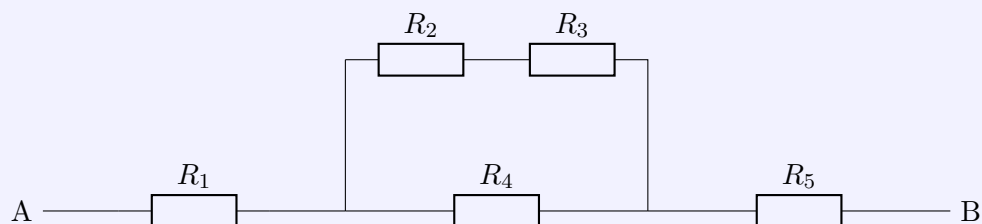
Three light bulbs are connected in series to a 120 V source as shown below. Find the current I through each of the bulbs. Each bulb is rated at 120 V. How much power is each bulb absorbing? Do they generate much light?

**Exercise 5: Equivalent Resistance 1**

Calculate the equivalent resistance R_{ab} across the terminals.

**Exercise 6: Equivalent Resistance 2**

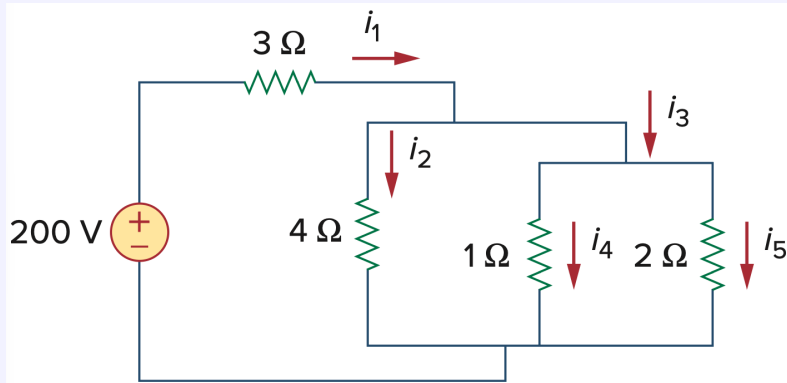
Given the resistor network with $R_1 = 10\ \Omega$, $R_2 = 20\ \Omega$, $R_3 = 30\ \Omega$, $R_4 = 5\ \Omega$, $R_5 = 15\ \Omega$:



Calculate the equivalent resistance of the network between terminals A and B.

Exercise 7: Current Divider Rule

Calculate i_1 to i_5 .



Exercise 8: Circuit Analysis

Calculate the equivalent resistance across the terminals.

